THE PARTICLE ON A RING

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Abstract. We study the quantum-mechanical system of a particle on a ring enclosing integral or half-integral flux. In the half-integral case, we observe a Pin group symmetry which enables symmetry-protected degenerate ground states which are not broken by instanton effects. Along the way, we will see hallmarks of topological physics, including symmetries with anomalies and Lagrangians with topological terms.

1. Introduction

It is sometimes the case that a classical symmetry of a system does not extend to the quantum theory. When this happens, the symmetry is said to have an *anomaly*. In this project, we will explore a quantum-mechanical model with an anomalous symmetry in order to elucidate the general behavior of anomalies, as symmetries with anomalies occur frequently in quantum field theory.

2. Projective Representations

To see how anomalies arise, consider first as an analogy a free spin-$\frac{1}{2}$ particle. Classically, a free particle has an $SO(3)$ 3-dimensional rotation symmetry. We can attempt to quantize this symmetry by introducing symmetry operators corresponding to each rotation in $SO(3)$ such that they form a *representation* of $SO(3)$ on the Hilbert space of states.

Definition 2.1. A *representation* is a map $\rho : G \to GL(V)$ from a group G to operators on a vector space V satisfying $\rho(gh) = \rho(g)\rho(h)$.

We employ the standard formula for rotation operators in this system in terms of the Pauli vector $\vec{\sigma}$.

$$R(\theta, \hat{n}) = \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right)(\hat{n} \cdot \vec{\sigma})$$

Notice that this quantization has an odd quirk: the same rotation seemingly corresponds to two different operators! In particular, if we do a 2π rotation to our system, we don't get back to where we started. We instead pick up a minus sign. Therefore, the map from $SO(3)$ to the symmetry operators isn't actually a function and so can't be a representation. This is an example of a classical symmetry of the system failing to become the system's quantum symmetry.

In quantum mechanics, this frequently happens because states are only well-defined up to global phase, so the classical consequences of the theory are independent of this phase factor. We can modify our definition of a representation to take this into account.

Definition 2.2. A *projective representation* is a map $\rho : G \to GL(V)$ from a group G to operators on a vector space V satisfying $\rho(gh) = e^{i\phi(g,h)}\rho(g)\rho(h)$ for some $\phi : G \to \mathbb{R}$. Equivalently, it is a group homomorphism $G \to PGL(V)$, or a representation *up to global phase*.

Our free spin-$\frac{1}{2}$ particle therefore has a projective representation of $SO(3)$ instead of an ordinary one. Mathematics have devised a trick to study projective representations.

Definition 2.3. A *central extension* of a group G by a group A is a group E such that

- $A \subseteq E$ and $E/A \simeq G$
- Every element of A commutes with every element of E.

Theorem 2.4. *Every projective representation of a group lifts to an ordinary representation of a central extension of the group.*

So projective representations are just ordinary representations in disguise! In the case of the spin-$\frac{1}{2}$, we can extend $SO(3)$ by the generator -1 to get the group $Spin(3)$, which does admit an ordinary representation and is therefore the actual quantum symmetry group. In particular, this group is isomorphic to $SU(2)$, the group of 2x2 unitary matrices with determinant 1. With these central extensions, we're able to compute exactly what symmetry the anomalous classical symmetry becomes in the quantum theory.

3. The Free Theory

Now that we understand how anomalies could occur, we will study a toy model of a system with an anomalous symmetry, following [1]. Consider a free particle moving on a unit circle enclosing a magnetic flux chosen such that the total Aharonov-Bohm phase of the particle moving once around the circle is θ. The corresponding Lagrangian is

$$L = \frac{1}{2}\dot{q}^2 + \frac{\theta}{2\pi}\dot{q}$$

The $\frac{\theta}{2\pi}\dot{q}$ term looks like a total derivative and so classically should not contribute to the Lagrangian. However, it is necessary for getting the right phase factor. From the path integral

$$\int \mathcal{D}q \, \exp(i \int dt \frac{1}{2}\dot{q}^2 + \frac{\theta}{2\pi}\dot{q}) = \int \mathcal{D}q \, \exp(iS_{\text{free}}) e^{i\theta \frac{\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}}}{2\pi}}$$

we see that $\frac{\theta}{2\pi}\dot{q}$ is necessary for the Aharonov-Bohm phase for paths where $\tilde{q}_{\text{final}} - \tilde{q}_{\text{initial}} = 2\pi k$. (We use $\tilde{q}$ to denote the continuous position variable which includes winding around the circle, so it may be greater than 2π.) It is an example of a *topological term*. Its contribution to the action depends only on the winding number of the path, which is a topological invariant.

A more subtle phenomenon is that θ and $\theta + 2\pi$ correspond to equivalent theories, since we can compute the path integral

$$\begin{aligned}\int \mathcal{D}q \, \exp(iS_{\text{free}}) e^{i(\theta+2\pi)\frac{\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}}}{2\pi}} &= \int \mathcal{D}q \, \exp(iS_{\text{free}}) e^{i\theta \frac{\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}}}{2\pi}} e^{i(\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}})} \\ &= e^{i(\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}})} \int \mathcal{D}q \, \exp(iS_{\text{free}}) e^{i\theta \frac{\tilde{q}_{\text{final}} - \tilde{q}_{\text{inital}}}{2\pi}}\end{aligned}$$

and find that they are equal up to a phase dependent only on the endpoints, which means the theories are equivalent. From the Lagrangian, we can construct the Hamiltonian

$$H = \frac{1}{2}\left(\hat{p}_q - \frac{\theta}{2\pi}\right)^2$$

and find that it is diagonal in the Fourier basis. Its eigenfunctions and eigenvalues are

$$\psi_n(q) = \frac{1}{\sqrt{2\pi}} e^{inq} \qquad E_n = \frac{1}{2}\left(n - \frac{\theta}{2\pi}\right)^2$$

For all values of θ, there is a rotational symmetry operator O_a corresponding to $q \mapsto q + a$. This acts on the energy eigenstates by

$$|n\rangle \mapsto e^{ian} |n\rangle$$

We can also consider a parity symmetry O_c, which acts by $q \mapsto -q$, $\theta \mapsto -\theta$. For $\theta = 0$, this is also a symmetry of the system. These generators satisfy

$$O_c O_a O_c \left|n\right\rangle = e^{-ian} \left|n\right\rangle = O_{-a} \left|n\right\rangle$$

The full group of symmetries is therefore $O(2)$, the group of rotations and reflections of a circle.

Interestingly, O_c is also a symmetry for $\theta = \pi$, since we can identify theories $\theta \sim \theta + 2\pi$. In this case, we have the unitary identification of states

$$\left|n\right\rangle_\theta \sim \left|n+1\right\rangle_{\theta+2\pi}$$

so the symmetry acts by

$$\left|n\right\rangle_\theta \mapsto \left|-n\right\rangle_{-\theta} \sim \left|-n+1\right\rangle_\theta$$

We can compute

$$O_c O_a O_c \left|n\right\rangle = e^{-ian+ia} \left|n\right\rangle = e^{ia} O_{-a} \left|n\right\rangle$$

and find that the relations of $O(2)$ are satified only up to a phase factor e^{ia}. Hence, the symmetries are only a *projective* representation of $O(2)$! Symmetry groups realized projectively like this are the heart of anomalies in quantum mechanics and quantum field theory.

As we have seen, this projective representation must lift to an ordinary representation of a central extension of $O(2)$. But which central extension? We consider the new generator $V_a = I_{-\frac{a}{2}} O_a$ for $a \in [0, 4\pi]$ and observe that it satisfies

$$O_c V_a O_c = V_{-a}$$

If we quotient by the two-element subgroup $\{V_0, V_{2\pi}\} = \{+1, -1\}$, this is just $O(2)$ again. Therefore, the group of symmetries generated by $\{O_c, V_a\}$ is a double cover of of $O(2)$. We call this group $Pin(2)$. Just as the *Spin* groups are double covers of $SO(n)$, the *Pin* groups are double covers of $O(n)$.

4. $\theta = 0$ with Interactions

We now subject this model to an external potential $V(\cos(2q))$. Note that this potential is invariant under both $q \to -q$ and $q \to q + \pi$. We will additionally assume that the potential has global minima at 0 and π. As a warm up, we first consider the $\theta = 0$ theory, which should be seen as representative of the generic case. This potential has two classical minima: one at 0 and one at π. We might then expect two ground states localized at 0 and π. However, quantum mechanics forbids this. Since symmetry operators must commute with the Hamiltonian, the ground state must respect the symmetries of the system. Therefore, we should have energy eigenstates akin to $\frac{1}{\sqrt{2}}(\left|0\right\rangle \pm \left|\pi\right\rangle)$, where $\left|0\right\rangle$ and $\left|\pi\right\rangle$ are localized around 0 and π.

We might expect these states to have the same energy. If we do perturbation theory around each minimum separately, that's exactly what we would find. However, the real behavior is more subtle than that. Recall that we can compute the energy spectrum using the Euclidean path integral.

$$\sum_n \psi_n^*(a)\psi_n(b) e^{-\frac{\beta E_n}{\hbar}} = \left\langle q = a\right| e^{\frac{-\beta H}{\hbar}} \left|q = b\right\rangle = \int \mathcal{D}q e^{-\frac{S_E}{\hbar}}$$

When working with the path integral, we expanded around the classical path. But which classical path should we take? Recall that for the free particle on the circle, we had to sum over all classical paths from one point to another, indexed by winding number. These topologically distinct paths all had different actions, but they were all stationary point of the action functional and so contributed significantly to the path integral. This model has a similar behavior in its Euclidean path integral. The leading order term in the path integral comes from expanding around the paths that are stationary at the classical minima of 0 and π. These do not split the degeneracy. However, there

are also solutions that tunnel in Euclidean time from 0 to π and locally minimize the Euclidean energy. These are called *instantons*.

Definition 4.1. *Instantons* are solutions to the classical equations of motion of a system in Euclidean time.

Since they locally minimize the Euclidean action, they have a significant contribution to the Euclidean path integral. Additionally, they contribute *nonlocal* effects, which can reveal phenomena not seen in perturbation theory. In particular, instantons can (and generically do) split the ground state degeneracy. The specifics of this splitting depend on the exact potential chosen, but for a double-well potential like $V(q) = -\cos(2q)$, for example, instanton effects make the energy of $\frac{1}{\sqrt{2}}(|0\rangle + |\pi\rangle)$ lower than the energy of $\frac{1}{\sqrt{2}}(|0\rangle - |\pi\rangle)$

5. $\theta = \pi$ with Interactions

We now consider instanton effects when $\theta = \pi$. The Euclidean action is then

$$S_E = \int d\tau \left(\frac{1}{2}\dot{q}^2 + V(q) - \frac{i}{2}\dot{q}) \right)$$

Notice something interesting: Since the potential is symmetric both under $q \to -q$ and $q \to q+\pi$, for every path q from 0 to π we can compute both

$$e^{-S_E(q)} = e^{-\int d\tau \left(\frac{1}{2}\dot{q}^2 + V(q)\right)} e^{\int d\tau \frac{i}{2}\dot{q}} = (\pm i) e^{-\int d\tau \left(\frac{1}{2}\dot{q}^2 + V(q)\right)}$$

$$e^{-S_E(-q)} = e^{-\int d\tau \left(\frac{1}{2}\dot{q}^2 + V(-q)\right)} e^{-\int d\tau \frac{i}{2}\dot{q}} = (\mp i) e^{-\int d\tau \left(\frac{1}{2}\dot{q}^2 + V(q)\right)}$$

so $e^{-S_E(q)} = -e^{-S_E(-q)}$. Therefore, the contributions from both paths to the path integral cancel, and the instanton effects vanish! In particular,

$$\sum_n \psi_n^*(\pi)\psi_n(0) e^{-\frac{\beta E_n}{\hbar}} = 0$$

Since the ground state is localized around 0 and π, $\psi^*(\pi)\psi(0) \neq 0$. Therefore, the ground state must have an unbroken degeneracy such that the contributions from each state to value of the path integral cancel. For example, there could be two ground state wave functions which satisfy $\psi_1^*(\pi)\psi_1(0) = -\psi_2^*(\pi)\psi_2(0)$, but we need at least two for the contributions to cancel.

Notice that this argument is independent of the actual potential, as long as it is symmetric under $q \to -q$ and $q \to q + \pi$. In fact, it is possible to see this degeneracy using only representation-theoretic considerations without any computation with path integrals. When we quantize the theory, the classical symmetries become symmetry operators O_c and O_π. Individually, both of these satisfy

$$O_c^2 = O_\pi^2 = 1$$

so each symmetry individually is anomaly-free. However, from Section 3, we know that

$$O_c O_\pi = -O_\pi O_c$$

so $O_c O_\pi$ is actually an element of order 4. Since this is not the case in the classical theory, we say the system has a *mixed anomaly* between O_c and O_π. We then identify the symmetry of the system with a central extension of $\mathbb{Z}_2 \times \mathbb{Z}_2$ known as D_8, a group of order 8. The operators therefore form a representation of D_8 on the Hilbert space. Additionally, since $[O_c, O_\pi] = -1$ acts nontrivially by scalar multiplication, we know that this representation is such that every subrepresentation is nonabelian. Therefore, there can be no one-dimensional energy eigenspaces, since otherwise the operators on that eigenspace would act only by multiplication by a phase, which is abelian. Thus, every energy eigenvalue must have a degeneracy.

6. Conclusion and Extensions

The toy model above has many analogies for more advanced concepts in topological physics, in particular ones involving actual quantum fields.

- *Terms that are sensitive to the topology of spacetime.* We saw this with the $\frac{\theta}{2}\dot{q}$ term in the Lagrangian, whose nontriviality depended on the particle being on a circle. One term that can appear in 2+1D gauge QFTs is a Chern-Simons term.
$$CS_k = \frac{k}{4\pi}\int_M tr\left(dA \wedge A + \frac{2}{3}A \wedge A \wedge A\right)$$
where M is spacetime. Under gauge transformations of the field A, the Chern-Simons term transforms according the winding number of the gauge transformation as a map $M \to G$, where G is the structure group. If $M = \mathbb{R}^3$, this winding number is always 0, but in other spacetimes this winding number can be nonzero, forcing quantization of the level k of the theory. This has consequences like the integer quantum Hall effect [2].
- *Anomalies.* Gaiotto et al. originally derived the model as an analogy for *'t Hooft anomalies* in gauge QFTs, which are obstructions to quantizing gauge theories. The ultimate cause of these is the same phase ambiguity that occurs when quantizing an classical symmetry in ordinary quantum mechanics, as seen by the model.
- *Symmetry protection.* In the model, we saw that the exact ground state degeneracy was robust against perturbations which preserve the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry. This *symmetry-protection* is common in condensed matter physics, especially *symmetry-protected topological phases*. For example, it has be seen experimentally that *topological insulators* are robust against perturbations which do not break time-reversal symmetry.

We therefore see how this model elucidates some of the unusual features of quantum field theories.

References

[1] Davide Gaiotto, Anton Kapustin, Zohar Komargodski, and Nathan Seiberg. Theta, Time Reversal, and Temperature. *JHEP*, 05:091, 2017.

[2] David Tong. The quantum hall effect, 2016.